

Gamified VR Educational Platform for Middle School Curriculum in Sri Lanka

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Introduction to the Problem



Research Problem

Traditional classroom methods are insufficient for delivering immersive, practical experiences in STEM education for grade 10 & 11 students.



Impact

- Disconnect between theory and practice among O/L students.
- Limited access to hands-on learning, especially in rural schools.
- Weak retention of ancient Sri Lankan innovations in younger generations.

Why we chose ?



STEM subjects often feel abstract and disengaging to students – especially in rural areas with limited lab or workshop facilities.



Virtual Reality (VR) enables immersive, hands-on experiences that simulate real-world processes – ideal for bridging this gap.



Many students struggle with Science, Mathematics, and Technology due to a lack of practical, relatable learning environments.



Our historical VR village connects students to Sri Lanka's rich engineering and agricultural heritage while teaching core curriculum content.



Blending culture with curriculum helps students understand STEM concepts in a familiar, story-driven context – especially appealing for visual and experiential learners.

Target Groups



Primary Users

Grade 10–11 students preparing for O/L exams.



Secondary Users

Teachers, content designers, rural education developers.



Students with limited lab/workshop access and prefer visual learning modes.



Social or Research Impact of the Solution



The solution promotes active learning through VR. Students interact with Science, Maths, Technology and Engineering in historical settings.



It raises motivation through gamification with points, missions, and rewards.



It supports bilingual delivery in Sinhala and English. This helps rural students.



It preserves cultural identity by embedding historical content.



It aligns with the national curriculum. Students gain practical support.



Previous Research and Shortcomings



Global VR apps exist but are not syllabus aligned. They focus on general learners.



Local projects were prototypes. They lacked integration with other subjects and had no bilingual support.



Existing tools often need internet access, which rural schools lack.



This project builds a syllabus aligned VR platform. It integrates all four subjects inside historical contexts. It includes bilingual narration and offline use.

Design Excellence of Our Solution



Cultural Relevance

Embedded Sinhala terminology, accurate architectural design.



Gamified Flow

Student's complete challenges tied to real-world knowledge.



Visual Design

Stylized historical models made using Blender & Unity.



Accessibility

Dual language support, low hardware requirement for schools.

Constraints and Limitations



Hardware access is limited to available Meta Quest devices.



Teacher training is required for classroom use.



Student readiness is uneven, especially in rural areas.



The project will begin with pilot testing in selected schools. Scaling will depend on results and resources.

Learning Mathematics in Gamified VR Educational Platform With History

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Specialization: Interactive Media



Proven Gap & Creative Solution



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Mathematics lessons in Sri Lankan schools are mostly taught using abstract problems and textbook exercises, making it difficult for students to connect concepts to real life. Topics such as measurement, fractions, algebra, and geometry are often memorized without proper understanding. There is a lack of interactive, visual, and context-based learning tools that help students apply mathematics in meaningful situations.

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The Lakmaga VR Mathematics module transforms abstract mathematics into interactive, real-world learning. Students explore a virtual ancient Sri Lankan environment where they measure land, calculate areas and volumes, apply ratios, and solve problems related to irrigation, construction, and trade. This approach makes mathematics practical, engaging, and culturally relevant

Research Gap

Research Area	Current Development	Our Mathematics Contribution
VR in Math Education	Mostly foreign syllabus-based	Aligned with Sri Lankan Grades 7–9 curriculum
Teaching Measurement	Paper-based problems only	Real-world measurement inside VR
Area & Volume Learning	Formula memorization	Hands-on calculation using VR objects
Algebra & Fractions	Abstract symbols	Visual and interactive problem-solving
Student Engagement	Low motivation	Gamified VR math challenges
Research Area	Current Development	Our Mathematics Contribution

Objectives

Main Objective



To design and develop a gamified Virtual Reality (VR) mathematics learning module that enhances conceptual understanding, spatial reasoning, and student engagement among Sri Lankan Grade 10-11 students.

Sub Objectives



- Teach accurate measurement and unit conversion
- Apply area, volume, and ratios to real scenarios
- Improve understanding of fractions, algebra, and geometry
- Develop logical thinking and problem-solving skills
- Reduce math anxiety through gamified learning



Progress



- Measurement tasks implemented in VR
- Area and volume calculation activities designed
- Ratio-based water distribution problems included
- Visual models for fractions and geometry added
- Step-by-step problem-solving logic introduced

Future Implements



- Adaptive difficulty for math challenges
- More real-world word problems
- Timed math missions with rewards
- In-module quizzes and progress tracking
- Teacher analytics and student performance logs



Functional Requirements

- Measurement Tasks: Measure length, width, and depth in VR
- Area & Volume Calculation: Use real objects and fields
- Fractions & Ratios: Divide land, water, and resources
- Algebra & Geometry: Solve equations using visual models
- Bilingual Support: Sinhala and English instructions
- Gamification: Points, badges, and levels
- Assessments: Quizzes after each math activity



Non-Functional Requirements

- Easy-to-use VR interface for middle-school students
- Stable performance with 72+ FPS
- Comfortable and safe VR movement
- Culturally accurate Sri Lankan environments
- Modular design for adding new math topics
- Compatible with Meta Quest headsets
- Offline functionality for rural schools



Design Excellence of Our Solution



Culturally Grounded

Uses local farming practices and ecosystems relevant to Sri Lankan students (traditional pest control).



Immersive Learning

Interactive VR environments simulate real-time farming, ecosystems, and sustainability scenarios.



Curriculum-Aligned

Modules reflect Grade 10–11 Science syllabus with local relevance and practical application.



Bilingual Accessibility

Sinhala and English voice guidance, labels, and narration enhance inclusivity.



Visual Engagement

Realistic crop models, dynamic weather, and animated food chains deepen conceptual understanding.

Learning Science in Gamified VR Educational Platform With History

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Specialization: Interactive Media



Proven Gap & Creative Solution



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Students-especially in rural Sri Lankan schools-lack practical exposure to scientific concepts like agriculture, ecosystems, and pest control. Current lessons are textbook-based, disengaging, and disconnected from real-life applications, with no localized or interactive tools available.

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The Lakmaga VR Science module turns theory into immersive learning. Students explore a virtual ancient village, interact with real-time crop cycles, control pests, and observe water systems like the Bisokotuwa - making science hands-on, engaging, and rooted in local context.

Research Gap

Research Name	Current Development (What they researched / where they stopped)	Future & Special Features (Our Module Contribution)
Virtual Labs in Science Teaching: Sri Lankan Teachers' Perspectives and Challenges	Found VR labs boost engagement; focused on teacher views, not classroom roll-out.	Deliver offline VR labs with Sinhala–English support, gamification, cross-subject links, and full syllabus alignment.
Transforming Science Education with VR: Immersive Representations Model (IRM)	Proposed an evaluation framework; no practical classroom implementation.	Build hands-on VR lessons with Sinhala–English UI/voice, gamified paths, cross-curricular tasks, and offline use.
Virtual Reality in Science Education: A Descriptive Review	Positive results mainly in higher ed; weak school-level curriculum integration.	Provide school-level, syllabus-mapped VR science with Sinhala–English support, gamification, and offline access.
Adding Immersive VR to a Science Lab Simulation (Presence↑, Learning↓)	VR raised presence but reduced learning vs desktop; lacks pedagogy fixes.	Improve outcomes via guided interactions, Sinhala–English explanations, analytics, and mission-based design.
Bridging Gaps in Science Education with Virtual Labs	Noted internet/teacher-training barriers; theory only.	Ship offline VR labs + Sinhala–English teacher guides, gamified student tasks, and curriculum alignment.

Objectives

Main Objective



To enhance students' understanding of scientific concepts in agriculture and ecosystems by providing an immersive and interactive VR-based learning experience rooted in Sri Lankan context.

Sub Objectives



- Simulate key agricultural processes (e.g., planting, pest control, crop cycles).
- Visualize scientific phenomena like the water cycle and photosynthesis.
- Provide practical science learning for students without lab access.
- Increase student engagement through gamified scientific tasks.



Progress



- Paddy Field Interaction implemented
- Pest Control Simulation working
- Ecosystem Module with Food Chain visuals
- Human Impact: Sustainable vs. Unsustainable practices

Future Implements



- Gamified progress tracking and rewards
- Improved food chain order selection
- Enhanced pest control realism
- Integrated in-module assessments
- Character animations and interactions
- Structure of Biosphere labelling



Functional Requirements

- Paddy Field Interaction: Plant, water, and observe crop growth stages.
- Pest Control Simulation: Identify and spray pests in real scenarios.
- Lifecycle Visualization: Animate germination to harvest stages.
- Bilingual Content: Sinhala & English narration and labels.
- Gamified Tracking: Points/rewards for task completion.
- Assessments: Quizzes and challenges after each module.



Non-Functional Requirements

- Usability & Comfort: Child-friendly controls, stable 72+ FPS, short load times, and VR comfort practices.
- Safety: Guardian boundaries, headset warnings, and safe-modeled lab practices.
- Localization: Incorporate Sri Lankan culture, fauna, flora, attire, and narration style.
- Maintainability & Scalability: Modular Unity design for updates, scalable to multiple headsets and future grades.
- Compatibility & Evaluation: Optimized for Meta Quest 3 (portable to Quest 2/other XR), with activity logging for research and debugging.



Design Excellence of Our Solution



Culturally Grounded

Uses local farming practices and ecosystems relevant to Sri Lankan students (traditional pest control).



Immersive Learning

Interactive VR environments simulate real-time farming, ecosystems, and sustainability scenarios.



Curriculum-Aligned

Modules reflect Grade 10–11 Science syllabus with local relevance and practical application.



Bilingual Accessibility

Sinhala and English voice guidance, labels, and narration enhance inclusivity.



Visual Engagement

Realistic crop models, dynamic weather, and animated food chains deepen conceptual understanding.

Learning Technology in Gamified VR Educational Platform With History

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Specialization: Interactive Media



Proven Gap & Creative Solution



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Many students, especially in rural Sri Lankan schools, do not get enough practical exposure to technology. Most lessons explain technology only through textbooks and pictures. Students rarely see how water systems, storage methods, and simple machines actually work. There are very few interactive and local learning tools that show how Sri Lankans used technology to solve real problems in the past.

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The Lakmaga VR Technology module turns these lessons into hands-on experiences. Students explore a virtual ancient Sri Lankan village, operate water tanks and spillways, observe food storage systems, and use simple machines like pulleys. This helps students learn technology in a practical, engaging, and local way.

Research Gap

Existing Research	What They Did	What Our Project Adds
VR in Technology Education	Focused on modern engineering only	Introduces ancient Sri Lankan technology
Digital Irrigation Lessons	Only diagrams and explanations	Interactive water flow and tank control
Learning Simple Machines	Videos and 2D images	Hands-on VR interaction
Sustainability Education	Mostly theory-based	Shows real sustainable systems
Rural Technology Learning	Limited by resources	Offline VR learning

Objectives

Main Objective



To help students understand technology and problem-solving by learning through a VR experience based on ancient Sri Lankan technology.

Sub Objectives



- Show how water tanks and irrigation systems work
- Explain the Bisokotuwa and water pressure control
- Demonstrate food storage and preservation methods
- Teach simple machines like pulleys and levers
- Show how technology supports sustainability and efficiency



Progress



- Tank structure and bund models completed
- Gamified VR tasks
- Granaries and storage pits created
- Environmental adaptation concepts shown

Future Implements



- Interactive Bisokotuwa controls
- Gamified challenges using machines
- In-module quizzes and tasks
- Guide characters for explanations



Functional Requirements

- Tank Interaction: Store and release water
- Bisokotuwa Simulation: Control water pressure
- Storage Systems: View and explore granaries
- Simple Machines: Lift and move objects
- Bilingual Content: Sinhala and English
- Gamification: Tasks and rewards
- Assessments: Short quizzes and challenges



Non-Functional Requirements

- Easy-to-use and comfortable VR controls
- Safe movement and boundary protection
- Sri Lankan buildings, tools, and culture
- Modular system for future updates
- Works on Meta Quest 3 and other headsets
- Activity logging for research evaluation



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Visual Engagement

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Learning Engineering in Gamified VR Educational Platform With History

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Specialization: Interactive Media



Proven Gap & Creative Solution



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Engineering concepts in Sri Lankan schools are mostly taught through textbooks and diagrams, giving students little chance to see how systems actually work. Topics such as water management, structural stability, and material selection are difficult to understand without practical experience. There are very few interactive and locally relevant engineering learning tools.

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The Lakmaga VR Engineering module provides hands-on learning through a virtual ancient Sri Lankan village. Students interact with water systems, explore structures, and identify engineering materials, helping them understand engineering concepts in a practical and meaningful way.

Research Gap

Research Area	Current Development	Our Engineering Contribution
VR in Engineering Education	Mostly focuses on modern or foreign contexts	Uses ancient Sri Lankan engineering systems
Teaching Water Engineering	Static diagrams and explanations	Interactive water flow and control simulations
Structural Engineering Education	Theory-based with few models	Hands-on structure exploration in VR
Materials Science Learning	Descriptive learning only	Material identification through VR interaction
School-Level Engineering Tools	Limited access to labs	Offline VR-based engineering practice

Objectives

Main Objective



To develop a VR-based gamified engineering education module set in a historical Sri Lankan village, aligned with the Grade 10–11 Engineering Technology curriculum, to improve student engagement and conceptual understanding through immersive, interactive learning.

Sub Objectives



- Increase student engagement and motivation in engineering subjects through gamified VR tasks.
- Improve visualization and understanding of abstract engineering concepts using 3D interaction.
- Align all learning activities with Sri Lankan Grade 10–11 engineering syllabus outcomes.
- Integrate culturally relevant historical contexts into engineering learning.



Progress



- Designed an engineering-focused VR task based on the foundation construction of a stupa
- Implemented a foundation-building activity where the player must place materials in the correct order
- Integrated a real-time feedback mechanism
- Gamified task progression

Future Implements



- Develop a new task focused on building the upper part of the stupa using bricks
- Implement visual and physical feedback
- Increased gamification
- System refinement and testing



Functional Requirements

- Provide an immersive 3D VR environment representing an ancient village
- Allow users to navigate freely using VR
- Enable interaction with virtual objects
- Provide real-time feedback on user actions (success, failure, hints).
- Implement gamification elements such as points, levels, badges, and task progression.



Non-Functional Requirements

- Maintain smooth VR performance with high frame rates to prevent motion sickness.
- Ensure the system is easy to use for Grade 10–11 students with minimal training.
- Support multiple languages and clear visual/audio instructions.
- System should run stably without crashes during classroom use.



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Commercialization & Market

Ability of Commercialization

- The VR platform is syllabus aligned for Grades 7–9 in Sri Lanka.
- Bilingual support (Sinhala and English) makes it accessible for rural and urban schools.
- Scalable across public and private schools, tuition centers, and NGOs.
- VR headsets are already piloted in Sri Lanka, showing readiness for adoption.



Investment / Cost and Recovery

Item	Estimated Cost (LKR)	Notes	Recovery / Pricing Strategy
Development per module (Math, Science, English, History)	5,000 – 10,000	Covers 3D assets, narration, testing, deployment	Cost recovered through annual school subscriptions
Total development (4 modules)	20,000 – 40,000	Full platform build excluding hardware	Breakeven within 2–3 years depending on adoption scale
Hardware (Meta Quest 3 headset)	~180,000	Primary classroom device, one per school initially	Shared across subjects, one-time investment
Other expenses (travel, teacher training, contingency)	~20,000 – 30,000	Pilot deployments and support	Covered through subscription margin or NGO funding
Suggested pricing	10,000 – 15,000 per school / year	Affordable for schools, scalable with adoption	Subscription and update fees ensure recurring revenue

Target Market, Customer Profile, Value Proposition, IPR

Category	Details
Target Market	Public and private middle schools in Sri Lanka (Grades 7–9). Tuition centers and NGOs focused on bilingual and STEM education.
Customer Profile	School administrators looking for modern, syllabus-aligned solutions. Teachers needing practical VR modules. Students seeking interactive, bilingual learning.
Value Proposition	Affordable VR platform aligned to national syllabus. Covers Math, Science, English, and History. Offers bilingual Sinhala-English support. Works offline for rural schools. Includes gamification for motivation.
Intellectual Property Rights	Ownership of custom 3D models, historical reconstructions, bilingual narration scripts, and VR gamification framework. Protectable via copyright and licensing for schools.



Thank You!

Gamified VR Educational Platform for Middle School Curriculum in Sri Lanka